

Where Extreme Motion Begins: Downstream-Return Seed Selection over Enumerated Self-Motion Branches

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Abstract. In extreme motion generation, a fixed-base manipulator maximizes the Cartesian path length along a pre-defined trajectory before any safety boundary is violated. Prior work has shown that the achievable length is largely decided before execution, by the choice of the initial joint configuration among the disconnected branches of the self-motion manifold. This paper studies that first decision as the opening macro-action of a two-stage semi-Markov decision process and asks four questions: whether the candidate set itself bounds the system, which supervision signal the selection requires, how much of the long-horizon outcome is predictable from static information, and whether the selection stage and the downstream controller must be adapted jointly. On 10,000-scale straight-line path-following tasks with a 7-DoF Franka FR3, we find that (i) replacing a learned generative proposal pool with direct cone-constrained IK enumeration more than doubles the attainable headroom; (ii) on the enumerated pool every geometric surrogate fails, whereas a selector supervised by complete closed-loop returns recovers about two thirds of the headroom; (iii) this fraction is a measured representability boundary: granting the selector a full kinematic-lookahead oracle adds nothing; (iv) forward controller adaptation is verified to be unnecessary here, and a novelty-stratified analysis explains why. The resulting system deploys as one static selector forward, one seed, and one rollout, and lengthens the average path by 24.5 mm and 27.7 mm over the strongest prior system on two held-out sets of 2,048 tasks each; the gain is stable across six training seeds and reaches +29.2 mm on a sealed 10,000-task final holdout.

Keywords: extreme motion generation, seed selection, reinforcement learning

1 Introduction

Consider again a compact, fixed-base manipulator executing a coating or welding operation along a straight line [1]. The motion is continuous and strictly constrained by safety boundaries such as joint limits and tool-orientation tolerance, and in practice one of these boundaries is violated long before the end-effector

approaches the edge of the reachable workspace. We refer to the problem of deferring these violations as *extreme motion generation*: maximizing the Cartesian path length along a pre-defined trajectory within the manipulator’s workspace.

Our previous study addressed the execution stage of this problem with a hybrid controller and initialized the motion by sampling a conditional diffusion model over favorable starting configurations [1]. The present paper turns to the initialization stage itself, motivated by a structural fact of redundant manipulation. For a task that constrains fewer degrees of freedom than the arm possesses, the feasible initial configurations form a self-motion manifold whose joint-limit-induced branches are disconnected; configurations on different branches realize the same end-effector pose yet differ sharply in how far the subsequent motion can proceed. Three properties make this initial choice scientifically distinctive. First, it is *decisive*: on our task distribution, the best and worst feasible starting configurations of the same task differ by a median of 472 mm of final path length, more than 85 % of the average task length. Second, its consequence is *revealed late*: whether a branch survives is dominated by orientation-cone violations that occur deep into the trajectory, and, as we show, is only weakly visible in any instantaneous geometric quantity at $t=0$. Third, the decision must be made *without trial*: an execution budget of one rollout per task excludes probing candidates, and because a full episode lasts only about 55 control steps, any informative probe would cost nearly as much as exhaustive execution. The initial choice must therefore compress the outcome of an entire closed-loop execution into a single static decision.

We formalize the initial configuration as the first, discrete macro-action of a two-stage semi-Markov decision process whose continuation is the continuous low-level controller, and whose shared objective is the final path length. This formulation admits two coupling directions in the sense of sequential policy chaining [2]: a *backward* direction, in which complete controller returns supervise the selection, and a *forward* direction, in which the controller adapts to the selection-induced initial-state distribution. Rather than assuming both directions are necessary, we treat their necessity as an experimental question. The contributions are four measured answers:

1. **The candidate set bounds the system.** Replacing the learned generative proposal pool of [1] with direct cone-constrained IK enumeration and farthest-point selection raises the per-task attainable ceiling by 52 mm on average and more than doubles the selectable headroom; the ceiling saturates by pool size $K \approx 16\text{--}24$, so enumeration coverage, not a learned generator, is what matters (Sec. 6.1).
2. **Backward credit is necessary, not auxiliary.** On the enumerated pool, every static geometric surrogate and every selector transferred from the generative pool performs *worse* than a trivial first-feasible rule, while the same feature set supervised by complete closed-loop returns recovers 62–68 % of the oracle headroom (Sec. 6.2).
3. **The static limit is measured, not conjectured.** Capture plateaus near two thirds under capacity, schedule, and architecture variation, and—decisively—

granting the selector an exact kinematic-lookahead feature, computed by integrating the damped tracking field to termination, leaves capture unchanged. The residual third of the headroom requires information about the learned controller’s own closed-loop behavior, which no static quantity can carry (Sec. 6.3).

4. **Forward adaptation is verified unnecessary here.** Two progressively better-engineered rounds of controller adaptation on the selection-induced distribution yield a null effect, the loop’s fail-closed gate rejecting one harmful and one neutral update. A novelty-stratified analysis shows the adaptation gain is zero even for seeds 3–8 rad away from any configuration the prior pool contained, and explains the mechanism: task-relative observations combined with training over 10^5 random tasks already cover the relative state space that new branches occupy (Sec. 6.4).

The deployed system keeps the strictest execution budget—one selector forward, one seed, one controller rollout, no probes—and lengthens the average path by 24.5 mm and 27.7 mm over the strongest prior system on two held-out sets of 2,048 tasks (95 % CIs [16.1, 33.0] and [18.5, 36.9]), with a 5 % trimmed gain of 14.6 mm and 18.0 mm confirming that typical tasks, not only tail cases, improve. The result is stable across six selector training seeds and reaches +29.2 mm (CI [24.8, 33.4]) on a sealed 10,000-task holdout generated after every design decision was frozen.

2 Related Work

2.1 Initialization of Redundant Manipulation

The influence of the initial configuration on downstream feasibility has long been recognized in redundancy resolution. Classical treatments optimize instantaneous criteria such as manipulability [3] or joint-limit distance, and modern planners embed these measures into sampling and cost [4,5]. Generative approaches model the full inverse-kinematics solution set, either with normalizing flows [6] or diffusion models [7], and DiffusionSeeder [8] seeds downstream motion with multi-modal trajectory initializations, closely related to the conditional sampler of our prior system [1]. All of these methods score or generate initializations from information available at $t=0$. Our experiments quantify exactly how far such information can go on a long-horizon objective: instantaneous surrogates fail outright on a diverse candidate set, learned static selection recovers two thirds of the headroom, and the remainder is shown to be invisible to any static feature, including an exact kinematic continuation.

2.2 Chaining and Bidirectional Policy Adaptation

Sequential Dexterity [2] fine-tunes adjacent sub-policies bidirectionally through a transition-feasibility function so that each skill terminates inside the basin

of its successor. Our two-stage decomposition borrows this credit structure—the first stage must be evaluated by the success of the second—but differs in what the experiments support. In our system the backward direction carries the entire gain, while the forward direction is verified to be already saturated by the controller’s design; we provide a falsifiable coverage-based explanation and delineate when forward adaptation should become necessary. We view this as complementary evidence: the value of each chaining direction is a property of the system, to be measured rather than assumed.

2.3 Learning from Complete Returns

Supervising a fast decision-maker with the outcome of an expensive full evaluation follows the pattern of expert iteration and amortized planning [9]. Here the expensive evaluation is the complete closed-loop rollout of every feasible candidate under a frozen controller, and the amortized decision is a listwise selector queried once per task. Unlike distillation from a search teacher, the complete return table is exhaustive over the discrete action set, so the selector’s residual error isolates the representability of the outcome, a property we exploit in Sec. 6.3.

3 Problem Formulation

We adopt the task setting of [1]. A straight-line path-following task is specified by the condition vector $\mathbf{c} = [\mathbf{p}_0, \mathbf{d}, \mathbf{n}] \in \mathbb{R}^9$, where $\mathbf{p}_0$ is the starting position and $\mathbf{d}, \mathbf{n}$ are unit vectors giving the motion direction and the task-plane normal. A 7-DoF manipulator with configuration $\mathbf{q} \in \mathbb{R}^7$ must keep its end-effector position $\mathbf{p}(\mathbf{q})$ within ϵ_{pos} of the advancing reference on the line and its tool axis $\mathbf{z}(\mathbf{q})$ within a cone of half-angle θ_{max} about $\mathbf{n}$, subject to joint limits; the episode terminates at the first violation, and the objective $\mathcal{L}$ is the net path length achieved before termination.

The two decisions of the system are made at different time scales, which we capture as a two-stage semi-Markov decision process. At $t=0$, a discrete macro-action selects the initial configuration from a finite feasible set $Q(\mathbf{c}) = \{\mathbf{q}_1, \dots, \mathbf{q}_K\}$ produced by a candidate generator; for $t>0$, the continuous controller π_{ctrl} acts at every step. Both stages share the terminal objective:

$$\max_{\phi, \theta} \mathbb{E}_{\mathbf{c}} [\mathcal{L}(\mathbf{q}_i, \pi_{\text{ctrl}, \theta}; \mathbf{c})], \quad i \sim \pi_{\text{seed}, \phi}(\cdot \mid \mathbf{c}, Q(\mathbf{c})). \quad (1)$$

Deployment is restricted to *one static pass*: the selector network is evaluated once, commits to a single $\mathbf{q}_0$, and the controller executes exactly one episode. No candidate probing, model-based rollout, or replanning is permitted at execution time. This restriction is not incidental: because episodes average only 55 steps, a probe long enough to reveal a branch’s fate costs nearly a full episode per candidate, collapsing any probing scheme into exhaustive execution. The scientific content of the problem is therefore whether, and to what extent, the terminal outcome of Eq. (1) can be predicted from information available before motion begins.

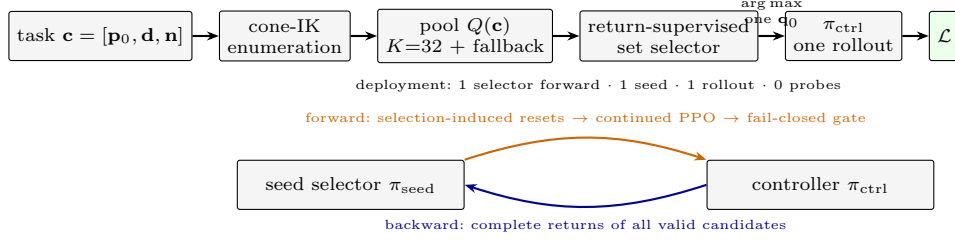


Fig. 1. The two-stage system. Top: the deployed path is one static selector forward over an enumerated cone-IK pool, followed by a single controller rollout. Bottom: the training loop transports terminal credit backward through complete candidate rollouts, and attempts forward controller adaptation on the selection-induced reset distribution behind a fail-closed promotion gate; on this task, the gate retains the original controller (Sec. 6.4).

4 Method

Figure 1 summarizes the system. We describe the three components in the order they answer the questions of Sec. 1.

4.1 Enumerated Candidate Pool

Instead of sampling a learned generative model, the candidate layer enumerates the feasible set directly. For each task, 128 damped-least-squares IK problems are solved from mixed random restarts toward tool directions sampled within the orientation cone, followed by joint-space deduplication at 0.08 rad; the median task yields 48 distinct solutions spread over the disconnected self-motion branches. Farthest-point selection in joint space then retains $K=32$ candidates, and the classical fallback configuration of [1] is appended as a designated final slot. Every candidate passes strict physical validation—joint limits, 5 mm position error, the $\theta_{\max}$ cone, and collision—before entering the action set; 94% of slots are valid on average. Diversity, not count, is the operative property: the attainable ceiling saturates by $K \approx 16$ –24 under farthest-point ordering (Sec. 6.1), and enumeration density rather than a learned prior determines coverage.

4.2 Backward Credit: Return-Supervised Selection

With the controller frozen, every valid candidate of every training task is rolled out to termination, producing a complete per-candidate return table. The selector is a five-member ensemble; each member embeds the K candidates’ 45-dimensional features—the controller’s initial observation augmented with the ray error, log positional manipulability, and ten task-aligned differential-kinematics descriptors [1]—with a permutation-equivariant encoder whose mean-pooled context is concatenated back to each candidate. Two heads are trained jointly:

a listwise scoring head fit to within-task range-normalized returns by softmax cross-entropy, and a feasibility head regressing the return in metres under a Huber loss. Members are trained on task-level bootstrap resamples and aggregated by mean score; deployment is the masked $\arg \max$ over valid candidates. The complete table, rather than sampled actions, is what makes the supervision a property of the *problem*: any residual selection error measures what the features cannot express, not what the data failed to visit.

4.3 Forward Adaptation under a Fail-Closed Gate

The forward direction freezes the selector and continues PPO training of the controller on the reset distribution it induces, mixing 70 % selector samples with 20 % uniform-valid and 10 % fallback resets to prevent collapse. Candidate controllers are admitted only through a promotion gate evaluated on paired held-out statistics; a rejected candidate leaves the deployed controller untouched. As Sec. 6.4 reports, both attempted rounds were declined—one as harmful, one as neutral—so the deployed system retains the original controller, and the gate’s audit trail becomes part of the scientific result.

5 Experimental Setup

All experiments use the straight-line path-following distribution of [1] with a simulated 7-DoF Franka FR3: tasks are drawn from a pool of 10^5 random lines, and the controller is the pure RL policy trained in that system. The prior system—the conditional diffusion proposal pool with its learned five-member ranker, deployed on the same controller—serves as the baseline throughout; we reuse its cached per-candidate rollouts so that both systems are compared on identical tasks under an identical controller.

Training uses 18,432 tasks with complete candidate rollouts (5.7×10^5 episodes). Model development decisions are made exclusively on a geometry-disjoint internal split (exact float32 task signatures; 3,000 held-out tasks); the two evaluation sets—the 2,048-task validation split and a 2,048-task external set built from a fresh line pool with audited zero geometry overlap—are read only for the results reported here. A sealed 10,000-task holdout was generated from fresh pool, task, and enumeration seeds after all rules, weights, and the selector publication rule (median across training seeds) were frozen and preregistered; it is read once, by the published selector only. We report paired per-task statistics against the baseline: the mean difference with a 20,000-resample bootstrap 95 % CI, the 5 % two-sided trimmed mean, and the fraction of tasks changed by more than ± 1 mm, alongside the fraction of the pool’s oracle headroom captured, $(\mathcal{L}_{\text{sel}} - \mathcal{L}_{\text{first}})/(\mathcal{L}_{\text{oracle}} - \mathcal{L}_{\text{first}})$.

Table 1. Candidate pools under the identical frozen controller (18,432 training tasks). Oracle = best valid candidate per task.

	generative pool ($K=8+1$)	enumerated pool ($K=32+1$)
valid candidates / task	7.9	31.0
first-feasible [m]	0.494	0.453
oracle [m]	0.582	0.634
headroom (oracle – first) [mm]	84.5	180.6
within-task spread, median [mm]	145	472

6 Results

6.1 The Candidate Pool Bounds the System

Table 1 and Fig. 2 compare the enumerated pool with the generative proposal pool on the 18,432 training tasks under the frozen controller. Enumeration raises the mean per-task ceiling from 0.582 m to 0.634 m (+52.1 mm; a better best exists on 68.4 % of tasks against 16.8 % for the converse) and widens the selectable headroom from 84.5 mm to 180.6 mm per task. The within-task spread between the best and worst feasible candidate (Fig. 3) shifts from a median of 145 mm to 472 mm: where the generative pool proposes near-duplicates of one good region, the enumerated pool spans branches whose outcomes differ by the scale of the task itself. The oracle curve saturates by $K \approx 16$ –24 under farthest-point ordering (98.4 % of the $K=32$ ceiling at $K=16$), so the benefit is branch coverage rather than candidate count. On the small fraction of tasks whose enumerated pool still lacks a competitor to the baseline’s choice (6–8 % of dev tasks), tripling the enumeration budget closes 71–73 % of the remaining gap and turns its mean sign negative, confirming an enumeration-coverage artifact rather than a manifold limitation; a learned generator is not required at any point.

6.2 Backward Credit is Necessary

Table 2 isolates the supervision signal on the enumerated pool. Every selection rule built from instantaneous information fails: the selector ensemble of the prior system, transferred unchanged, captures –25 % of the headroom (worse than taking the first feasible candidate), and the strongest single geometric feature—a component of the damped line-tracking joint velocity, with a within-task rank correlation of only 0.28—captures –23 %. The identical 45-dimensional feature set, retrained with complete closed-loop returns as targets, captures 62.7 % at 15,000 training tasks (66.6 % for the member ensemble), with a per-seed standard deviation of 0.2 percentage points. Figure 4 traces the supervision from 1,000 tasks upward; even a plain ridge regression on the same features reaches 23.6 %, confirming that the signal enters through the labels, not the model class. On this pool, downstream-return supervision is the difference between failing and working, which elevates backward credit transport from a refinement to a requirement.

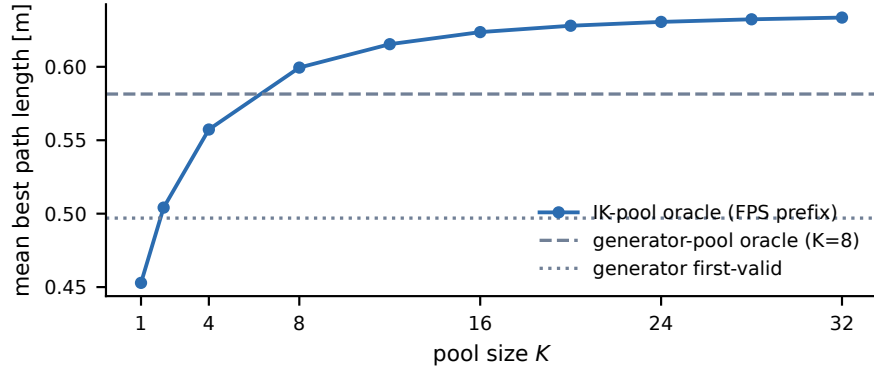


Fig. 2. Attainable ceiling versus pool size. The enumerated-pool oracle (farthest-point prefix) passes the full generative-pool oracle by $K=4$ and saturates by $K \approx 16-24$; marginal value at the tail is 0.2 mm per four candidates.

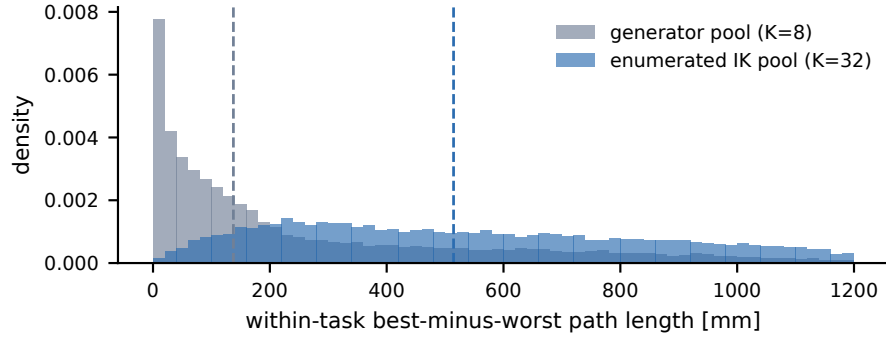


Fig. 3. Within-task best-minus-worst final path length. The enumerated pool (median 472 mm) exposes outcome diversity that the generative pool (median 145 mm) compresses away.

Table 2. Selection rules on the identical enumerated pool (held-out tasks; capture of oracle headroom).

selection rule	supervision	capture [%]
first feasible	none	0.0
best single geometric feature	none	-23.1
prior-system selector, transferred	generative-pool returns	-25.8
ridge on 45-D features	complete returns	+23.6
listwise selector (single)	complete returns	$+62.7 \pm 0.2$
listwise selector (ensemble)	complete returns	+66.6

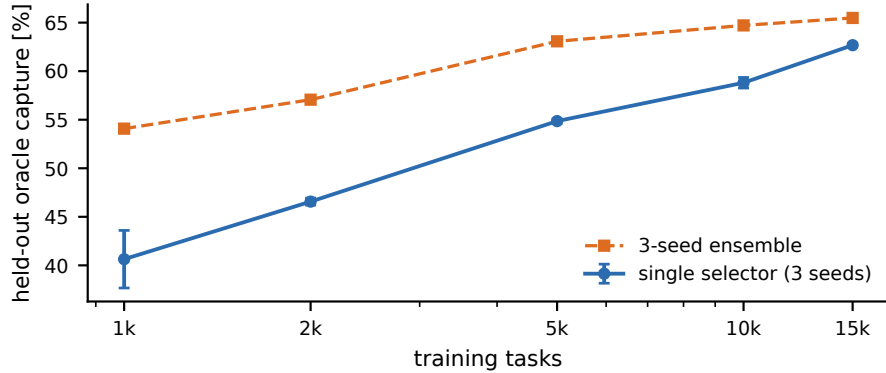


Fig. 4. Held-out capture versus training-set size (geometry-disjoint internal split). Seed-level scatter collapses as data grows; capacity and schedule variations at the right end change nothing (Sec. 6.3).

6.3 The Static Representability Boundary

Two thirds of the headroom is also where static prediction stops. Widening the network, doubling its depth of training, and exchanging the encoder all leave held-out capture at the same plateau (58.6–61.9 % single, 62.9–66.6 % ensemble). The decisive experiment grants the selector the strongest static information that exists: for every candidate we integrate the damped line-tracking field on the pure kinematic model until a constraint is met—a full nonlinear continuation of the starting branch, requiring no environment step and no policy call. This *kinematic lookahead* is individually excellent: its within-task rank correlation with the true outcome is 0.55, matching the entire trained network (0.53), and its arg max alone captures 52 % of the headroom. Yet appended to the 45 features and retrained, it moves capture by nothing (62.3 ± 0.9 single, 66.5 ensemble): the network had already internalized the kinematic continuation from its one-step descriptors. The plateau is therefore a measured boundary, not an engineering shortfall. What the boundary excludes is identified by the lookahead itself: the kinematic field survives 0.27 m on average where the learned controller achieves 0.415 m, and this behavioral surplus—how the policy spends the null space that no fixed field anticipates—is precisely the component of the outcome that static selection cannot see and that only execution reveals.

6.4 Forward Adaptation is Already Saturated

The forward direction was attempted twice under the promotion gate. A naive continuation of PPO on the selection-induced resets degraded the external set significantly (-2.75 mm, CI $[-4.7, -1.1]$) and was rejected. A properly-engineered second round—critic-only warm-up, zero entropy bonus, a tightened KL trust region, and a fourfold budget—was harmless but null (validation $+2.2$, CI $[-0.1, +4.6]$);

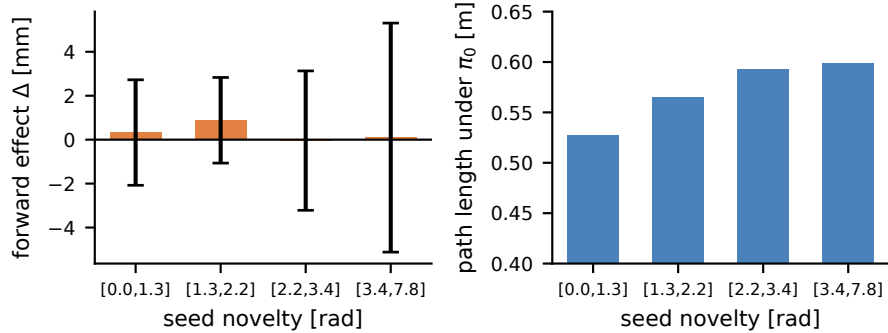


Fig. 5. Novelty-stratified test of forward adaptation (both dev sets, 4,096 tasks). Left: adaptation effect by quartile of joint-space distance from the selected seed to the nearest prior-pool configuration; all quartiles are null. Right: the *unadapted* controller’s path length rises with novelty—the selector seeks distant branches because they are better, and the controller already exploits them.

external -1.5 , CI $[-4.0, +1.0]$), with the training reward saturated at 0.99 throughout: under its own objective the controller had nothing left to learn from these states. Completing the loop regardless—relabeling all 5.7×10^5 candidate roll-outs under the adapted controller and retraining the selector—left every cell of the resulting 2×2 decomposition statistically tied, so the loop converges to its first iterate.

Figure 5 explains why, and makes the explanation falsifiable. Stratifying tasks by the joint-space distance between the newly selected seed and the nearest configuration available to the prior pool, the selector is seen to move far across the manifold: the median chosen seed lies 2.16 rad (90th percentile 4.61 rad) from anything the generative pool offered. If the controller harbored a coverage gap, adaptation gains would concentrate in the most novel stratum; instead the forward effect is zero in every quartile (top quartile $+0.09 \pm 5.2$ mm), and the unadapted controller performs *best* from the most novel branches (0.599 m versus 0.528 m in the least novel quartile). The mechanism is the product of three design choices inherited from [1]: task-relative observations, training over 10^5 random tasks, and reset mixtures containing uniform feasible configurations. In the relative state space the policy actually observes, an unfamiliar branch of one task is a familiar state of another, so the distribution shift that forward adaptation would repair has been absorbed in advance. Conversely, this delineates where the forward direction should matter: small task pools, absolute observations, or transfer to physical hardware.

6.5 System Comparison

Table 3 reports the deployed system—enumerated pool, published return-supervised selector, original controller, one static pass—against the prior system on identical

Table 3. Deployed system versus the prior system (identical tasks and controller; paired statistics; positive favors ours).

	validation (2,048)	external (2,048)	sealed (10,000)
prior system [m]	0.5384	0.5518	0.5443
ours [m]	0.5629	0.5795	0.5734
Δ mean [mm]	+24.5	+27.7	+29.2
95 % CI [mm]	[16.1, 33.0]	[18.5, 36.9]	[24.8, 33.4]
Δ trimmed 5 % [mm]	+14.6	+18.0	+16.8
win / harm >1 mm [%]	51.8 / 37.0	50.4 / 39.0	50.4 / 38.3
oracle capture [%]	67.2	68.9	67.3

Capture is measured against each pool’s own complete-candidate oracle; the enumerated pool’s oracle is itself 52 mm higher (Table 1).

tasks. The mean path lengthens by 24.5 mm (validation) and 27.7 mm (external), roughly fifteen times the margin the prior system’s own selector iteration achieved over its geometric predecessor, and the 5 % trimmed means of +14.6 and +18.0 mm show the gain is carried by typical tasks rather than a heavy tail. Figure 6 displays the full paired distribution: 51 % of tasks improve by more than 1 mm while 38 % regress—the enumerated pool’s weak first-feasible anchor means selection errors are costlier than in the generative pool, and a risk-sensitive selection objective is the natural next lever. Across six independent selector training seeds the gain is 25.5 ± 1.7 mm (validation) and 26.9 ± 1.5 mm (external)¹, and on the sealed 10,000-task holdout, read once with the published selector, the gain is +29.2 mm (CI [24.8, 33.4], trimmed +16.8 mm, capture 67.3 %). Deployment cost is unchanged where it matters: the selector forward remains sub-millisecond per task batched, candidate enumeration adds roughly half a second of offline kinematics per task, and the single controller rollout continues to dominate end-to-end latency.

6.6 Audit of the Bidirectional Loop

Because every update passed through preregistered gates, the loop leaves a complete decision trail: the first backward round was promoted (+24.5 / +27.7 mm); forward round one was rejected as harmful; forward round two was declined as null; the second backward round, trained on relabeled returns, tied its predecessor (−0.3 mm) and was not promoted; capacity, schedule, and lookahead-feature extensions were each declined on the internal split. Two further negative results complete the record. A margin gate that falls back to the classical seed raised harm instead of lowering it (37 % $\rightarrow$ 50 %), because the enumerated pool’s fall-back is itself weak; safe abstention requires a strong anchor, which this pool intentionally does not contain. And the loop’s convergence to its first iterate

¹Run seeds {0, 41000, . . . , 45000} under the preregistered protocol; the frozen median rule published run 0, which is also the deployed selector.

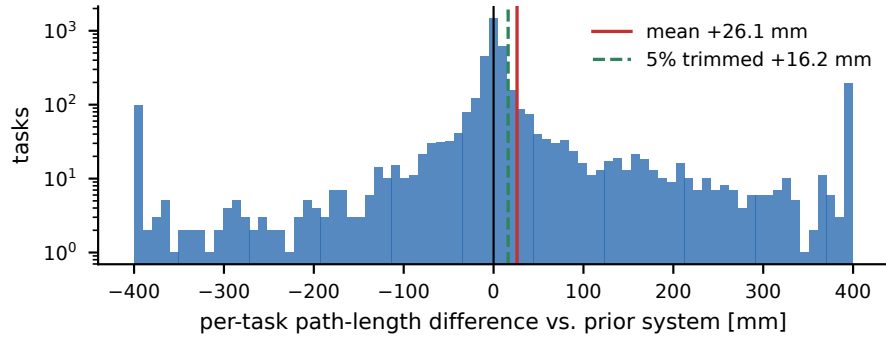


Fig. 6. Per-task paired differences against the prior system (both dev sets; log-scaled counts; clipped at ± 400 mm for display). The trimmed mean remains strongly positive: the improvement is not a tail artifact.

is not a failure of the framework but its verdict: on this system, the coupling between initialization and control lives entirely in the initialization.

7 Limitations

The remaining third of the headroom is shown to require execution-time information, and we do not offer a static route to it; whether an extremely short learned probe could price that information below oracle cost is open. Thirty-eight percent of tasks regress relative to the prior system despite the strongly positive mean and trimmed mean; a risk-sensitive selection objective is unexplored. All findings are in simulation under a single task family, and the coverage argument of Sec. 6.4 predicts—but does not verify—that forward adaptation becomes necessary on physical hardware.

8 Conclusion

We studied the opening decision of extreme motion generation as the first macro-action of a two-stage SMDP and measured, rather than assumed, each property of the problem: the candidate set was the binding constraint until enumeration replaced generation; complete downstream returns are the only supervision under which static selection works at all; static prediction ends, measurably, at two thirds of the headroom; and the forward coupling direction is already saturated by the controller’s design. The deployed system converts these answers into a 24–28 mm average extension of the achievable path at an unchanged execution budget of one seed and one rollout, confirmed across training seeds and a sealed holdout. We believe the same measurement discipline—treating each direction of a modular pipeline’s coupling as a hypothesis with a preregistered gate—transfers beyond this task.

References

1. Xinyi Yuan, Weiwei Wan, and Kensuke Harada. Extreme motion generation for straight-line path following via reinforcement learning. In *International Symposium on Robotics Research (ISRR)*, 2026.
2. Yuanpei Chen, Chen Wang, Li Fei-Fei, and C. Karen Liu. Sequential dexterity: Chaining dexterous policies for long-horizon manipulation. In *Proceedings of the 7th Conference on Robot Learning (CoRL)*, volume 229, pages 3809–3829. PMLR, 2023.
3. Tsuneo Yoshikawa. Manipulability and redundancy control of robotic mechanisms. In *Proceedings. 1985 IEEE International Conference on Robotics and Automation*, volume 2, pages 1004–1009. IEEE, 1985.
4. Henghua Shen, Wen-Fang Xie, Jianyu Tang, and Tao Zhou. Adaptive manipulability-based path planning strategy for industrial robot manipulators. *IEEE/ASME transactions on mechatronics*, 28(3):1742–1753, 2023.
5. Kuanqi Cai, Liding Zhang, Xinwen Su, Kejia Chen, Chaoqun Wang, Sami Haddadin, Alois Knoll, Arash Ajoudani, and Luis Figueredo. Just-in-time informed trees: Manipulability-aware asymptotically optimized motion planning. *IEEE/ASME Transactions on Mechatronics*, 2025.
6. Barrett Ames, Jeremy Morgan, and George Konidaris. Ikflow: Generating diverse inverse kinematics solutions. *IEEE Robotics and Automation Letters*, 7(3):7177–7184, 2022.
7. Zeyu Zhang and Ziyuan Jiao. Ikdiffuser: a diffusion-based generative inverse kinematics solver for kinematic trees. *arXiv preprint arXiv:2506.13087*, 2025.
8. Huang Huang, Balakumar Sundaralingam, Arsalan Mousavian, Adithyavairavan Murali, Ken Goldberg, and Dieter Fox. Diffusionseeder: Seeding motion optimization with diffusion for rapid motion planning. In *8th Annual Conference on Robot Learning*, 2024.
9. Xinyi Yuan, Weiwei Wan, and Kensuke Harada. Iksel: Selecting good seed joint values for iterative model-based ik solving, 2026. preprint.